Project Report

Devin Stuelund, Janet Lopez, Patrick Hopes

IS 4420

Company Description:

Best Buy is a leading retailer of consumer electronics, appliances, and technology services, operating through both physical stores and an online platform. This database focuses specifically on sales within the United States. In addition to product sales, Best Buy offers a range of services, including installation, repairs, and technical support through its Geek Squad division. A core objective of the company is to maintain high levels of customer satisfaction by providing reliable products and comprehensive support services.

User Requirements:

Geek Squad:
- Product Schematics
- Product Descriptions
- Service History
- Repair History
- Track Warranty Status

Salespeople:
- Available Products
- Customer Complaints
- Customer Resolutions
- Customer Purchase History

Managers:
- Inventory Management
- Tracking Employees' Sales

Customer Support:
- Order History
- Customer Demographics
- Returns and Refunds
- Delivery Report

Accounting/Finance:
- Inventory
- Past Sales
- Loss Report

Product
- Product Ordering
- Product Category
- Product Name

Business Rules:

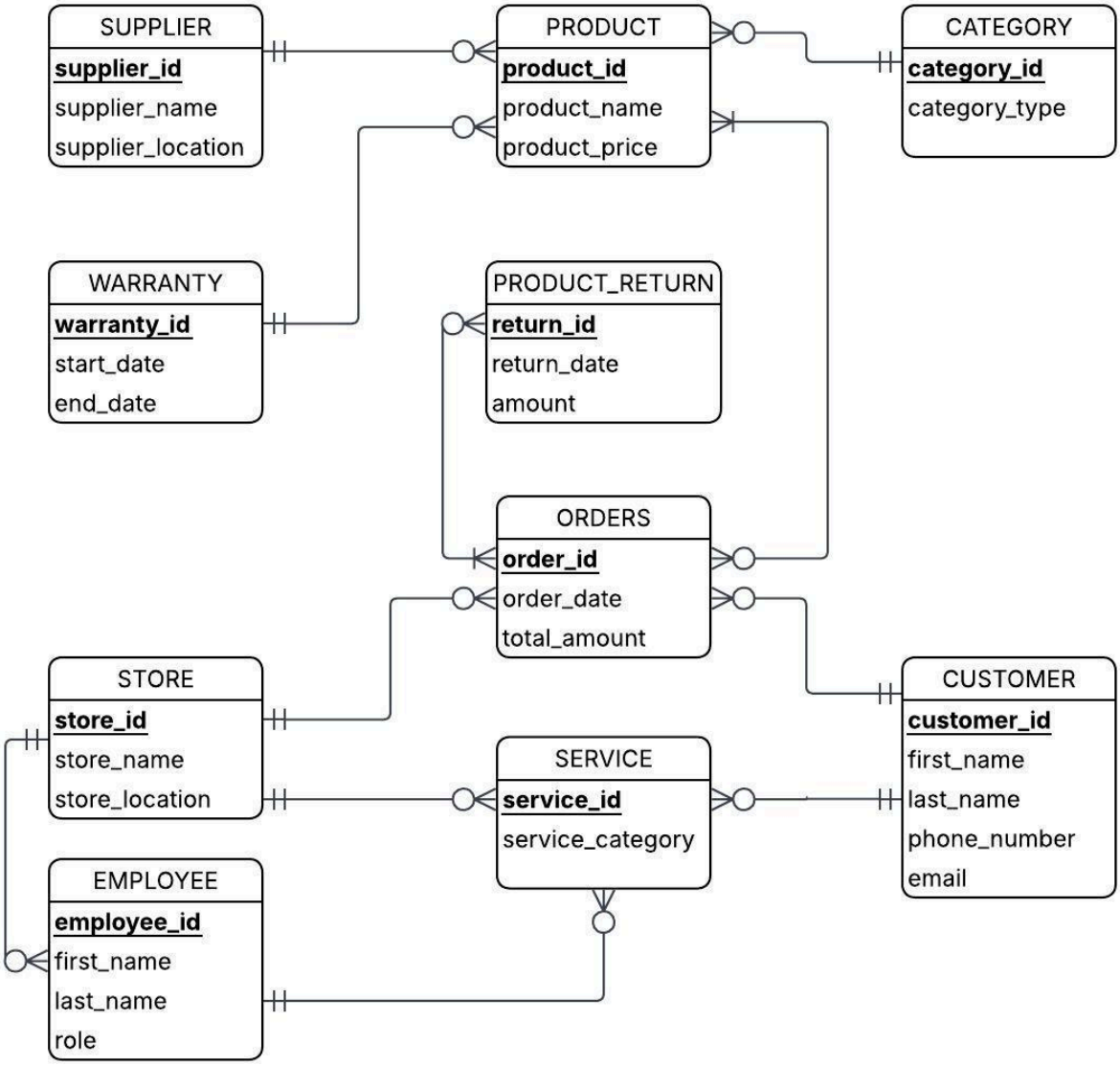
- One customer may place many orders, but one order must only be placed by one customer
- Each store may employ many employees, but each employee only works at one store

- Each store has at least one employee
- Each warranty may be associated with many products, but each product is associated with only one warranty
- Each supplier may provide many products, but each product is associated with one supplier
- Each store may process many orders, but each order is processed by one store
- One product category may include many products, but each product belongs to one category
- Each customer may request multiple services, but each service record belongs to one customer
- One employee may perform many service records, but each service record is performed by one employee
- One store may handle many service records, but each service record is handled by one store
- One product may have many return records, and each return record is associated with exactly one product
- A customer may have many returns, and each return is associated with one customer through the order it references
- Each product can be sold in at least one store, and each store can sell at least one product (its saying to remove this)
- An order needs at least one product, and multiple products can belong to one order
- One order may have many return records, but each return record belongs to one order

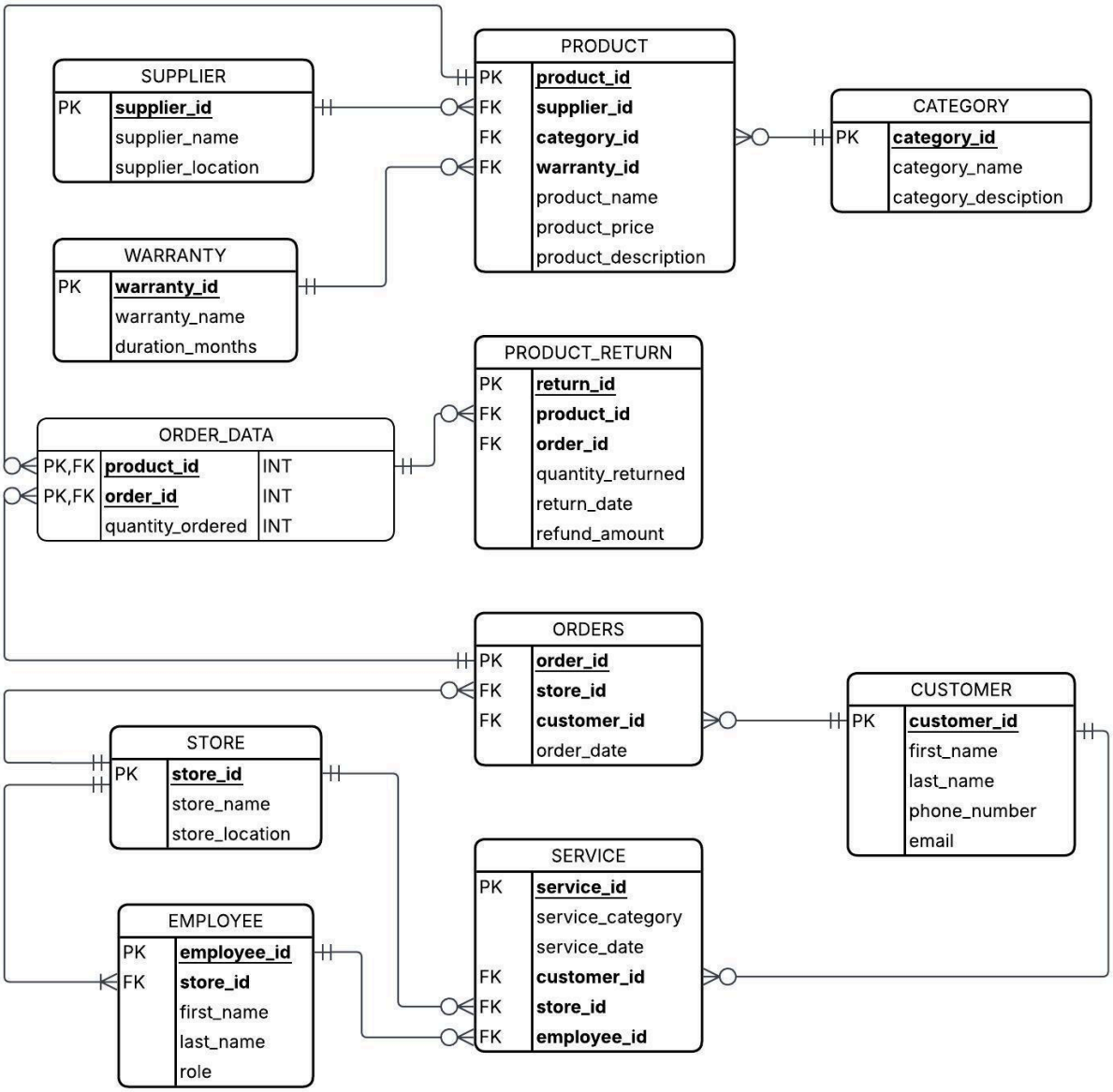
Data Outputs / Business Questions:

- Customer order report
- **Store sales report**
- Customer Satisfaction Report
- Inventory reports showing products in low stock
- Employee performance
- **Customer return and exchange reports**
- Reports showing average sales per employee
- Warranty service reports showing which products need the most repairs
- Monthly and yearly reports
- Store comparison reports showing which locations generate the highest sales

ERD
Conceptual Model:



Relational Model:



DATABASE SQL CODE (DDL)

Tables:

Supplier Table:

```sql
CREATE TABLE SUPPLIER (
    supplier_id INT NOT NULL,
    supplier_name VARCHAR(150) NOT NULL,
    supplier_location VARCHAR(150) NOT NULL,
    PRIMARY KEY (supplier_id)
);

DESC SUPPLIER;
```

100% 15:12

Result Grid | Filter Rows: Search | Export:

Field	Type	Null	Key	Default	Extra
supplier_id	int	NO	PRI	NULL	
supplier_name	varchar(150)	NO		NULL	
supplier_location	varchar(150)	NO		NULL	

Result 1 — Read Only

Warranty Table:

```sql
CREATE TABLE WARRANTY (
    warranty_id INT NOT NULL,
    warranty_name VARCHAR(150) NOT NULL,
    duration_months INT NOT NULL,
    PRIMARY KEY (warranty_id)
);

DESC WARRANTY;
```

100% 15:21

Result Grid | Filter Rows: Search | Export:

Field	Type	Null	Key	Default	Extra
warranty_id	int	NO	PRI	NULL	
warranty_name	varchar(150)	NO		NULL	
duration_months	int	NO		NULL	

Result 2 — Read Only

Category Table:

```
23  CREATE TABLE CATEGORY (
24      category_id INT NOT NULL,
25      category_name VARCHAR(150) NOT NULL,
26      category_description VARCHAR(255),
27      PRIMARY KEY (category_id)
28  );
29
30  DESC CATEGORY;
31
```

100% 15:30

Result Grid | Filter Rows: Search | Export:

Field	Type	Null	Key	Default	Extra
category_id	int	NO	PRI	NULL	
category_name	varchar(150)	NO		NULL	
category_description	varchar(255)	YES		NULL	

Result 3 Read Only

Store Table:

```
32  CREATE TABLE STORE (
33      store_id INT NOT NULL,
34      store_name VARCHAR(200) NOT NULL,
35      store_location VARCHAR(150) NOT NULL,
36      PRIMARY KEY (store_id)
37  );
38
39  DESC STORE;
40
```

100% 12:39

Result Grid | Filter Rows: Search | Export:

Field	Type	Null	Key	Default	Extra
store_id	int	NO	PRI	NULL	
store_name	varchar(200)	NO		NULL	
store_location	varchar(150)	NO		NULL	

Result 5 Read Only

Customer Table:

```
41  CREATE TABLE CUSTOMER (
42      customer_id INT NOT NULL,
43      first_name VARCHAR(150) NOT NULL,
44      last_name VARCHAR(150) NOT NULL,
45      phone_number VARCHAR(20) NOT NULL,
46      email VARCHAR(150) NOT NULL,
47      PRIMARY KEY (customer_id)
48  );
49
50  DESC CUSTOMER;
51
```

100% 15:50

Result Grid | Filter Rows: Search | Export:

Field	Type	Null	Key	Default	Extra
customer_id	int	NO	PRI	NULL	
first_name	varchar(150)	NO		NULL	
last_name	varchar(150)	NO		NULL	
phone_number	varchar(20)	NO		NULL	
email	varchar(150)	NO		NULL	

Result 6 Read Only

Employee Table:

```
    CREATE TABLE EMPLOYEE (
53      employee_id INT NOT NULL,
54      store_id INT NOT NULL,
55      first_name VARCHAR(150) NOT NULL,
56      last_name VARCHAR(150) NOT NULL,
57      role VARCHAR(150) NOT NULL,
58      PRIMARY KEY (employee_id),
59      FOREIGN KEY (store_id) REFERENCES STORE(store_id)
60  );
61
62  DESC EMPLOYEE;
63
```

100% 15:62

Result Grid | Filter Rows: Search | Export:

Field	Type	Null	Key	Default	Extra
employee_id	int	NO	PRI	NULL	
store_id	int	NO	MUL	NULL	
first_name	varchar(150)	NO		NULL	
last_name	varchar(150)	NO		NULL	
role	varchar(150)	NO		NULL	

Result 7 Read Only

Product Table:

```
CREATE TABLE PRODUCT (
    product_id INT NOT NULL,
    supplier_id INT NOT NULL,
    category_id INT NOT NULL,
    warranty_id INT NOT NULL,
    product_name VARCHAR(300) NOT NULL,
    product_price DECIMAL(10,2) NOT NULL,
    product_description TEXT,
    PRIMARY KEY (product_id),
    FOREIGN KEY (supplier_id) REFERENCES SUPPLIER(supplier_id),
    FOREIGN KEY (category_id) REFERENCES CATEGORY(category_id),
    FOREIGN KEY (warranty_id) REFERENCES WARRANTY(warranty_id)
);

DESC PRODUCT;
```

100% 14:78

Result Grid | Filter Rows: Search | Export:

Field	Type	Null	Key	Default	Extra
product_id	int	NO	PRI	NULL	
supplier_id	int	NO	MUL	NULL	
category_id	int	NO	MUL	NULL	
warranty_id	int	NO	MUL	NULL	
product_name	varchar(300)	NO		NULL	
product_price	decimal(10,2)	NO		NULL	
product_description	text	YES		NULL	

Result 8 Read Only

Order Table:

```
CREATE TABLE ORDERS (
    order_id INT NOT NULL,
    store_id INT NOT NULL,
    customer_id INT NOT NULL,
    order_date DATE NOT NULL,
    PRIMARY KEY (order_id),
    FOREIGN KEY (store_id) REFERENCES STORE(store_id),
    FOREIGN KEY (customer_id) REFERENCES CUSTOMER(customer_id)
);

DESC ORDERS;
```

100% 13:90

Result Grid | Filter Rows: Search | Export:

Field	Type	Null	Key	Default	Extra
order_id	int	NO	PRI	NULL	
store_id	int	NO	MUL	NULL	
customer_id	int	NO	MUL	NULL	
order_date	date	NO		NULL	

Result 9 Read Only

Service Table:

```
92  CREATE TABLE SERVICE (
93      service_id INT NOT NULL,
94      service_category VARCHAR(150) NOT NULL,
95      service_date DATE NOT NULL,
96      customer_id INT NOT NULL,
97      store_id INT NOT NULL,
98      employee_id INT NOT NULL,
99      PRIMARY KEY (service_id),
100     FOREIGN KEY (customer_id) REFERENCES CUSTOMER(customer_id),
101     FOREIGN KEY (store_id) REFERENCES STORE(store_id),
102     FOREIGN KEY (employee_id) REFERENCES EMPLOYEE(employee_id)
103 );
104
105 DESC SERVICE;
106
```

100% 14:105

Result Grid | Filter Rows: Search | Export:

Field	Type	Null	Key	Default	Extra
service_id	int	NO	PRI	NULL	
service_category	varchar(150)	NO		NULL	
service_date	date	NO		NULL	
customer_id	int	NO	MUL	NULL	
store_id	int	NO	MUL	NULL	
employee_id	int	NO	MUL	NULL	

Result 10 Read Only

Order Data Table:

```
107 CREATE TABLE ORDER_DATA (
108     product_id INT NOT NULL,
109     order_id INT NOT NULL,
110     quantity_ordered INT NOT NULL,
111     PRIMARY KEY (product_id, order_id),
112     FOREIGN KEY (product_id) REFERENCES PRODUCT(product_id),
113     FOREIGN KEY (order_id) REFERENCES ORDERS(order_id)
114 );
115
116 DESC ORDER_DATA;
117
```

100% 17:116

Result Grid | Filter Rows: Search | Export:

Field	Type	Null	Key	Default	Extra
product_id	int	NO	PRI	NULL	
order_id	int	NO	PRI	NULL	
quantity_ordered	int	NO		NULL	

Return Table:

```
118  CREATE TABLE RETURN_TABLE (
119      return_id INT NOT NULL,
120      product_id INT NOT NULL,
121      order_id INT NOT NULL,
122      quantity_returned INT NOT NULL,
123      return_date DATE NOT NULL,
124      return_amount DECIMAL(10,2) NOT NULL,
125      PRIMARY KEY (return_id),
126      FOREIGN KEY (product_id) REFERENCES PRODUCT(product_id),
127      FOREIGN KEY (order_id) REFERENCES ORDERS(order_id)
128  );
129
130  DESC RETURN_TABLE;
```

Export recordset to an external file

100% 19:130

Result Grid Filter Rows: Search Export:

Field	Type	Null	Key	Default	Extra
return_id	int	NO	PRI	NULL	
product_id	int	NO	MUL	NULL	
order_id	int	NO	MUL	NULL	
quantity_returned	int	NO		NULL	
return_date	date	NO		NULL	
return_amount	decimal(10,2)	NO		NULL	

Result Grid Form Editor

Foreign Keys:

Product:

```
131
132  ALTER TABLE PRODUCT
133  ADD CONSTRAINT fk_product_supplier
134  FOREIGN KEY (supplier_id) REFERENCES SUPPLIER(supplier_id);
135
136  SHOW CREATE TABLE PRODUCT;
```

100% 27:136

Result Grid Filter Rows: Search Export:

Table	Create Table
PRODUCT	CREATE TABLE `PRODUCT` (`product_id` int NOT NULL, `supplier_id` int NOT NULL, `category_id` int NOT NULL, `warranty_id` int NOT NULL, `pr…

Employee:

```
138  ALTER TABLE EMPLOYEE
139  ADD CONSTRAINT fk_employee_store
140  FOREIGN KEY (store_id) REFERENCES STORE(store_id);
141
142  SHOW CREATE TABLE EMPLOYEE;
```

100% 28:142

Result Grid Filter Rows: Search Export:

Table	Create Table
EMPLOYEE	CREATE TABLE `EMPLOYEE` (`employee_id` int NOT NULL, `store_id` int NOT NULL, `first_name` varchar(150) NOT NULL, `last_name` varchar(15…

Result Grid Form Editor

Order:

```
144 •  ALTER TABLE ORDERS
145    ADD CONSTRAINT fk_orders_customer
146    FOREIGN KEY (customer_id) REFERENCES CUSTOMER(customer_id);
147
148 •  ALTER TABLE ORDERS
149    ADD CONSTRAINT fk_orders_store
150    FOREIGN KEY (store_id) REFERENCES STORE(store_id);
151
152 •  SHOW CREATE TABLE ORDERS;
```

100% 26:152

Result Grid Filter Rows: Search Export:

Table	Create Table
ORDERS	CREATE TABLE `ORDERS` (`order_id` int NOT NULL, `store_id` int NOT NULL, `customer_id` int NOT NULL, `order_date` date NOT NULL, PRIMA…

Order Data:

```
154 •  ALTER TABLE ORDER_DATA
155    ADD CONSTRAINT fk_orderdata_order
156    FOREIGN KEY (order_id) REFERENCES ORDERS(order_id);
157
158 •  ALTER TABLE ORDER_DATA
159    ADD CONSTRAINT fk_orderdata_product
160    FOREIGN KEY (product_id) REFERENCES PRODUCT(product_id);
161
162 •  SHOW CREATE TABLE ORDER_DATA;
```

100% 30:162

Result Grid Filter Rows: Search Export:

Table	Create Table
ORDER_DATA	CREATE TABLE `ORDER_DATA` (`product_id` int NOT NULL, `order_id` int NOT NULL, `quantity_ordered` int NOT NULL, PRIMARY KEY (`product_i…

Service:

```
168 •  ALTER TABLE SERVICE
169    ADD CONSTRAINT fk_service_employee
170    FOREIGN KEY (product_id) REFERENCES PRODUCT(product_id);
171
172 •  ALTER TABLE SERVICE
173    ADD CONSTRAINT fk_service_store
174    FOREIGN KEY (store_id) REFERENCES STORE(store_id);
175
176 •  SHOW CREATE TABLE SERVICE;
177
```

100% 27:176

Result Grid Filter Rows: Search Export:

Table	Create Table
SERVICE	CREATE TABLE `SERVICE` (`service_id` int NOT NULL, `service_category` varchar(150) NOT NULL, `service_date` date NOT NULL, `custome

Return:

```
178 •  ALTER TABLE RETURN_TABLE
179    ADD CONSTRAINT fk_return_order
180    FOREIGN KEY (order_id) REFERENCES ORDERS(order_id);
181
182 •  ALTER TABLE RETURN_TABLE
183    ADD CONSTRAINT fk_return_product
184    FOREIGN KEY (product_id) REFERENCES PRODUCT(product_id);
185
186 •  SHOW CREATE TABLE RETURN_TABLE;
187
```

100% 32:186

Result Grid Filter Rows: Search Export:

Table	Create Table
RETURN_TABLE	CREATE TABLE `RETURN_TABLE` (`return_id` int NOT NULL, `product_id` int NOT NULL, `order_id` int NOT NULL, `quantity_returned

INSERT DATA INTO TABLES

supplier:

```
159
160 •  INSERT INTO `supplier` (`supplier_id`, `supplier_name`, `supplier_location`) VALUES
161     (1, 'Apex Components Ltd', 'Seattle, WA'),
162     (2, 'Northline Distributors', 'Portland, OR'),
163     (3, 'BluePeak Electronics', 'San Jose, CA'),
164     (4, 'Summit Tech Wholesale', 'Denver, CO'),
165     (5, 'Evergreen Consumer Supply', 'Austin, TX'),
166     (6, 'MetroParts Trading', 'Chicago, IL'),
167     (7, 'Horizon Appliance Supply', 'Atlanta, GA'),
168     (8, 'Quantum Logistics Supply', 'Dallas, TX'),
169     (9, 'Meridian Tech Imports', 'Newark, NJ'),
170     (10, 'Beacon Product Group', 'Phoenix, AZ'),
171     (11, 'Prime Source Solutions', 'Nashville, TN'),
172     (12, 'SilverLeaf Hardware', 'Charlotte, NC'),
173     (13, 'Nova Retail Partners', 'Miami, FL'),
174     (14, 'Vertex Global Supply', 'San Diego, CA'),
175     (15, 'Urban Circuit Supply', 'Los Angeles, CA');
176
177 •  SELECT * FROM supplier;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

supplier_id	supplier_name	supplier_location
1	Apex Components Ltd	Seattle, WA
2	Northline Distributors	Portland, OR
3	BluePeak Electronics	San Jose, CA
4	Summit Tech Wholesale	Denver, CO
5	Evergreen Consumer Supply	Austin, TX
6	MetroParts Trading	Chicago, IL
7	Horizon Appliance Supply	Atlanta, GA
8	Quantum Logistics Supply	Dallas, TX
9	Meridian Tech Imports	Newark, NJ
10	Beacon Product Group	Phoenix, AZ
11	Prime Source Solutions	Nashville, TN
12	SilverLeaf Hardware	Charlotte, NC
13	Nova Retail Partners	Miami, FL
14	Vertex Global Supply	San Diego, CA
15	Urban Circuit Supply	Los Angeles, CA
NULL	NULL	NULL

category:

```
178
179 • INSERT INTO `category` (`category_id`, `category_name`, `category_description`) VALUES
180     (1, 'Laptops', 'Portable computers for work, study, and everyday productivity.'),
181     (2, 'Tablets', 'Touchscreen devices for mobile browsing, media, and note-taking.'),
182     (3, 'Audio', 'Headphones, speakers, and sound accessories for personal and home use.'),
183     (4, 'Networking', 'Routers, mesh systems, and connectivity hardware for home offices.'),
184     (5, 'Displays', 'Monitors and screens for productivity, gaming, and retail setups.'),
185     (6, 'Peripherals', 'Input devices and desk accessories for computers and workstations.'),
186     (7, 'Storage', 'Solid-state drives, memory cards, and backup devices.'),
187     (8, 'Wearables', 'Smartwatches and fitness devices for health and productivity.'),
188     (9, 'Home Appliances', 'Small appliances and household convenience products.'),
189     (10, 'Smart Home', 'Connected devices for home automation and monitoring.'),
190     (11, 'Office Equipment', 'Printers, scanners, and related office hardware.'),
191     (12, 'Gaming', 'Gaming accessories and performance-focused hardware.'),
192     (13, 'Mobile Accessories', 'Cases, chargers, docks, and phone accessories.'),
193     (14, 'Power Accessories', 'Power banks, adapters, and charging solutions.'),
194     (15, 'Cables and Adapters', 'Data and power cables for modern device connectivity.');
195
196 • SELECT * FROM category;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

category_id	category_name	category_description
1	Laptops	Portable computers for work, study, and every...
2	Tablets	Touchscreen devices for mobile browsing, medi...
3	Audio	Headphones, speakers, and sound accessories ...
4	Networking	Routers, mesh systems, and connectivity hard...
5	Displays	Monitors and screens for productivity, gaming, ...
6	Peripherals	Input devices and desk accessories for compute...
7	Storage	Solid-state drives, memory cards, and backup d...
8	Wearables	Smartwatches and fitness devices for health an...
9	Home Appliances	Small appliances and household convenience pr...
10	Smart Home	Connected devices for home automation and m...
11	Office Equipment	Printers, scanners, and related office hardware.
12	Gaming	Gaming accessories and performance-focused h...
13	Mobile Accessories	Cases, chargers, docks, and phone accessories.
14	Power Accessories	Power banks, adapters, and charging solutions.
15	Cables and Ada...	Data and power cables for modern device conn...
NULL	NULL	NULL

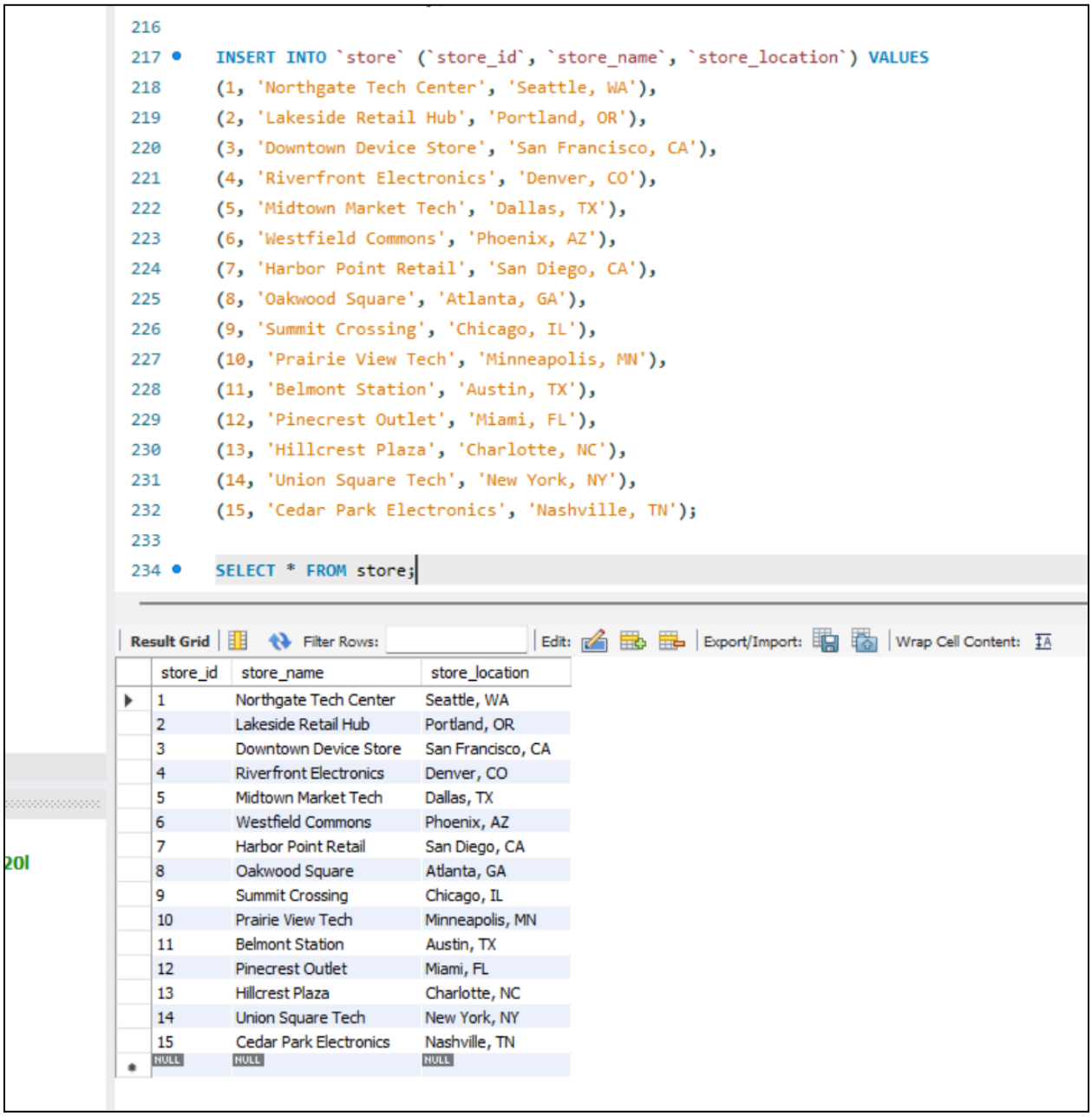
warranty:

```
197
198 • INSERT INTO `warranty` (`warranty_id`, `warranty_name`, `duration_months`) VALUES
199     (1, 'Standard Coverage', 12),
200     (2, 'Extended Coverage', 24),
201     (3, 'Premium Protection', 36),
202     (4, 'Accidental Damage Plan', 12),
203     (5, 'Business Protection', 24),
204     (6, 'Device Care Plus', 18),
205     (7, 'Replacement Plan', 6),
206     (8, 'Appliance Care', 24),
207     (9, 'Audio Care', 12),
208     (10, 'Smart Home Care', 18),
209     (11, 'Storage Shield', 36),
210     (12, 'Office Care', 24),
211     (13, 'Gaming Care', 24),
212     (14, 'Mobile Care', 12),
213     (15, 'Power Care', 12);
214
215 • SELECT * FROM warranty;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

warranty_id	warranty_name	duration_months
1	Standard Coverage	12
2	Extended Coverage	24
3	Premium Protection	36
4	Accidental Damage Plan	12
5	Business Protection	24
6	Device Care Plus	18
7	Replacement Plan	6
8	Appliance Care	24
9	Audio Care	12
10	Smart Home Care	18
11	Storage Shield	36
12	Office Care	24
13	Gaming Care	24
14	Mobile Care	12
15	Power Care	12
NULL	NULL	NULL

store:



```
216
217 • INSERT INTO `store` (`store_id`, `store_name`, `store_location`) VALUES
218   (1, 'Northgate Tech Center', 'Seattle, WA'),
219   (2, 'Lakeside Retail Hub', 'Portland, OR'),
220   (3, 'Downtown Device Store', 'San Francisco, CA'),
221   (4, 'Riverfront Electronics', 'Denver, CO'),
222   (5, 'Midtown Market Tech', 'Dallas, TX'),
223   (6, 'Westfield Commons', 'Phoenix, AZ'),
224   (7, 'Harbor Point Retail', 'San Diego, CA'),
225   (8, 'Oakwood Square', 'Atlanta, GA'),
226   (9, 'Summit Crossing', 'Chicago, IL'),
227   (10, 'Prairie View Tech', 'Minneapolis, MN'),
228   (11, 'Belmont Station', 'Austin, TX'),
229   (12, 'Pinecrest Outlet', 'Miami, FL'),
230   (13, 'Hillcrest Plaza', 'Charlotte, NC'),
231   (14, 'Union Square Tech', 'New York, NY'),
232   (15, 'Cedar Park Electronics', 'Nashville, TN');
233
234 • SELECT * FROM store;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

store_id	store_name	store_location
1	Northgate Tech Center	Seattle, WA
2	Lakeside Retail Hub	Portland, OR
3	Downtown Device Store	San Francisco, CA
4	Riverfront Electronics	Denver, CO
5	Midtown Market Tech	Dallas, TX
6	Westfield Commons	Phoenix, AZ
7	Harbor Point Retail	San Diego, CA
8	Oakwood Square	Atlanta, GA
9	Summit Crossing	Chicago, IL
10	Prairie View Tech	Minneapolis, MN
11	Belmont Station	Austin, TX
12	Pinecrest Outlet	Miami, FL
13	Hillcrest Plaza	Charlotte, NC
14	Union Square Tech	New York, NY
15	Cedar Park Electronics	Nashville, TN
NULL	NULL	NULL

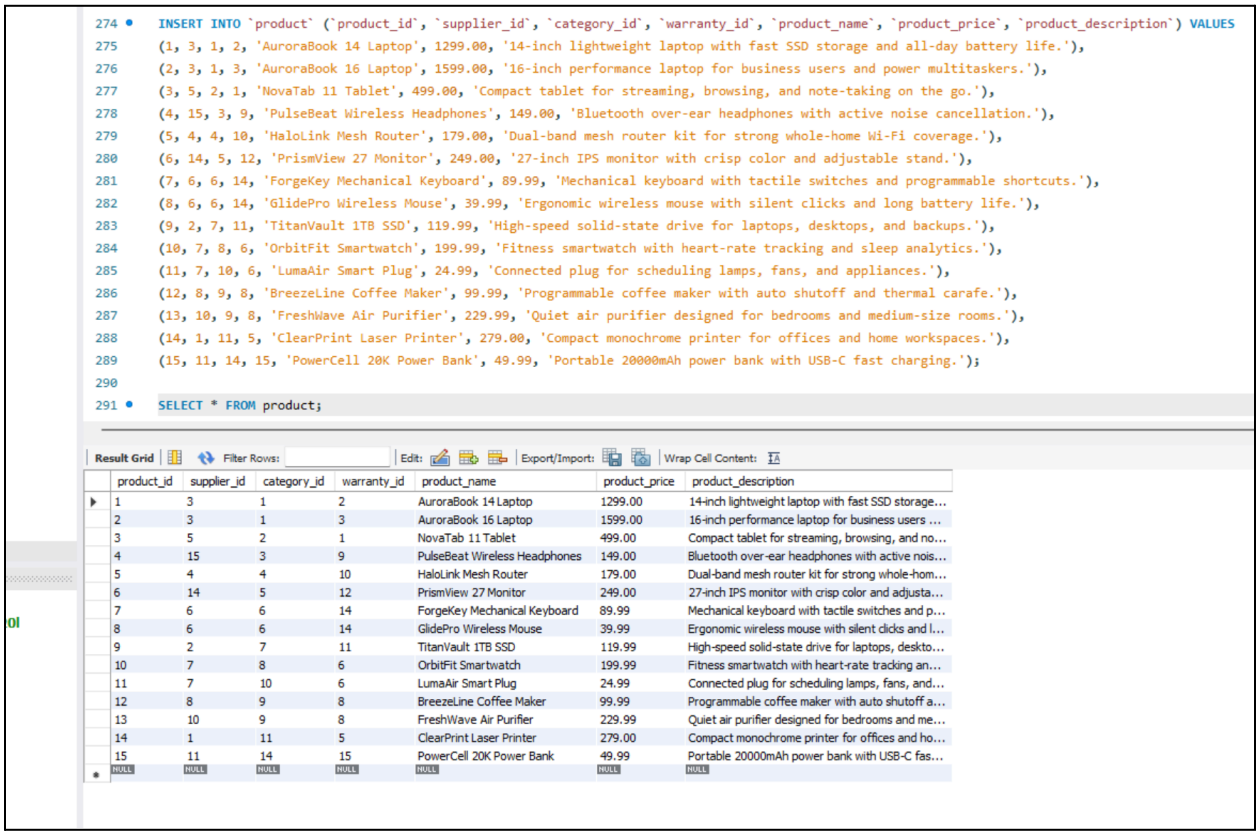
customer:

```
235
236 • INSERT INTO `customer` (`customer_id`, `first_name`, `last_name`, `phone_number`, `email`) VALUES
237   (1, 'Olivia', 'Reed', '206-555-0111', 'olivia.reed@example.com'),
238   (2, 'Ethan', 'Brooks', '503-555-0112', 'ethan.brooks@example.com'),
239   (3, 'Maya', 'Patel', '415-555-0113', 'maya.patel@example.com'),
240   (4, 'Noah', 'Carter', '303-555-0114', 'noah.carter@example.com'),
241   (5, 'Sophia', 'Nguyen', '214-555-0115', 'sophia.nguyen@example.com'),
242   (6, 'Liam', 'Mitchell', '602-555-0116', 'liam.mitchell@example.com'),
243   (7, 'Ava', 'Jackson', '619-555-0117', 'ava.jackson@example.com'),
244   (8, 'Lucas', 'Turner', '404-555-0118', 'lucas.turner@example.com'),
245   (9, 'Emma', 'Walker', '312-555-0119', 'emma.walker@example.com'),
246   (10, 'Jackson', 'Hughes', '612-555-0120', 'jackson.hughes@example.com'),
247   (11, 'Isabella', 'Foster', '512-555-0121', 'isabella.foster@example.com'),
248   (12, 'Mason', 'Price', '305-555-0122', 'mason.price@example.com'),
249   (13, 'Amelia', 'Ross', '704-555-0123', 'amelia.ross@example.com'),
250   (14, 'Logan', 'Bell', '212-555-0124', 'logan.bell@example.com'),
251   (15, 'Harper', 'Ward', '615-555-0125', 'harper.ward@example.com');
252
253 • SELECT * FROM customer;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

customer_id	first_name	last_name	phone_number	email
1	Olivia	Reed	206-555-0111	olivia.reed@example.com
2	Ethan	Brooks	503-555-0112	ethan.brooks@example.com
3	Maya	Patel	415-555-0113	maya.patel@example.com
4	Noah	Carter	303-555-0114	noah.carter@example.com
5	Sophia	Nguyen	214-555-0115	sophia.nguyen@example.com
6	Liam	Mitchell	602-555-0116	liam.mitchell@example.com
7	Ava	Jackson	619-555-0117	ava.jackson@example.com
8	Lucas	Turner	404-555-0118	lucas.turner@example.com
9	Emma	Walker	312-555-0119	emma.walker@example.com
10	Jackson	Hughes	612-555-0120	jackson.hughes@example.com
11	Isabella	Foster	512-555-0121	isabella.foster@example.com
12	Mason	Price	305-555-0122	mason.price@example.com
13	Amelia	Ross	704-555-0123	amelia.ross@example.com
14	Logan	Bell	212-555-0124	logan.bell@example.com
15	Harper	Ward	615-555-0125	harper.ward@example.com
NULL	NULL	NULL	NULL	NULL

product:



```
274 • INSERT INTO `product` (`product_id`, `supplier_id`, `category_id`, `warranty_id`, `product_name`, `product_price`, `product_description`) VALUES
275   (1, 3, 1, 2, 'AuroraBook 14 Laptop', 1299.00, '14-inch lightweight laptop with fast SSD storage and all-day battery life.'),
276   (2, 3, 1, 3, 'AuroraBook 16 Laptop', 1599.00, '16-inch performance laptop for business users and power multitaskers.'),
277   (3, 5, 2, 1, 'NovaTab 11 Tablet', 499.00, 'Compact tablet for streaming, browsing, and note-taking on the go.'),
278   (4, 15, 3, 9, 'PulseBeat Wireless Headphones', 149.00, 'Bluetooth over-ear headphones with active noise cancellation.'),
279   (5, 4, 4, 10, 'HaloLink Mesh Router', 179.00, 'Dual-band mesh router kit for strong whole-home Wi-Fi coverage.'),
280   (6, 14, 5, 12, 'PrismView 27 Monitor', 249.00, '27-inch IPS monitor with crisp color and adjustable stand.'),
281   (7, 6, 6, 14, 'ForgeKey Mechanical Keyboard', 89.99, 'Mechanical keyboard with tactile switches and programmable shortcuts.'),
282   (8, 6, 6, 14, 'GlidePro Wireless Mouse', 39.99, 'Ergonomic wireless mouse with silent clicks and long battery life.'),
283   (9, 2, 7, 11, 'TitanVault 1TB SSD', 119.99, 'High-speed solid-state drive for laptops, desktops, and backups.'),
284   (10, 7, 8, 6, 'OrbitFit Smartwatch', 199.99, 'Fitness smartwatch with heart-rate tracking and sleep analytics.'),
285   (11, 7, 10, 6, 'LumaAir Smart Plug', 24.99, 'Connected plug for scheduling lamps, fans, and appliances.'),
286   (12, 8, 9, 8, 'BreezeLine Coffee Maker', 99.99, 'Programmable coffee maker with auto shutoff and thermal carafe.'),
287   (13, 10, 9, 8, 'FreshWave Air Purifier', 229.99, 'Quiet air purifier designed for bedrooms and medium-size rooms.'),
288   (14, 1, 11, 5, 'ClearPrint Laser Printer', 279.00, 'Compact monochrome printer for offices and home workspaces.'),
289   (15, 11, 14, 15, 'PowerCell 20K Power Bank', 49.99, 'Portable 20000mAh power bank with USB-C fast charging.');
290
291 • SELECT * FROM product;
```

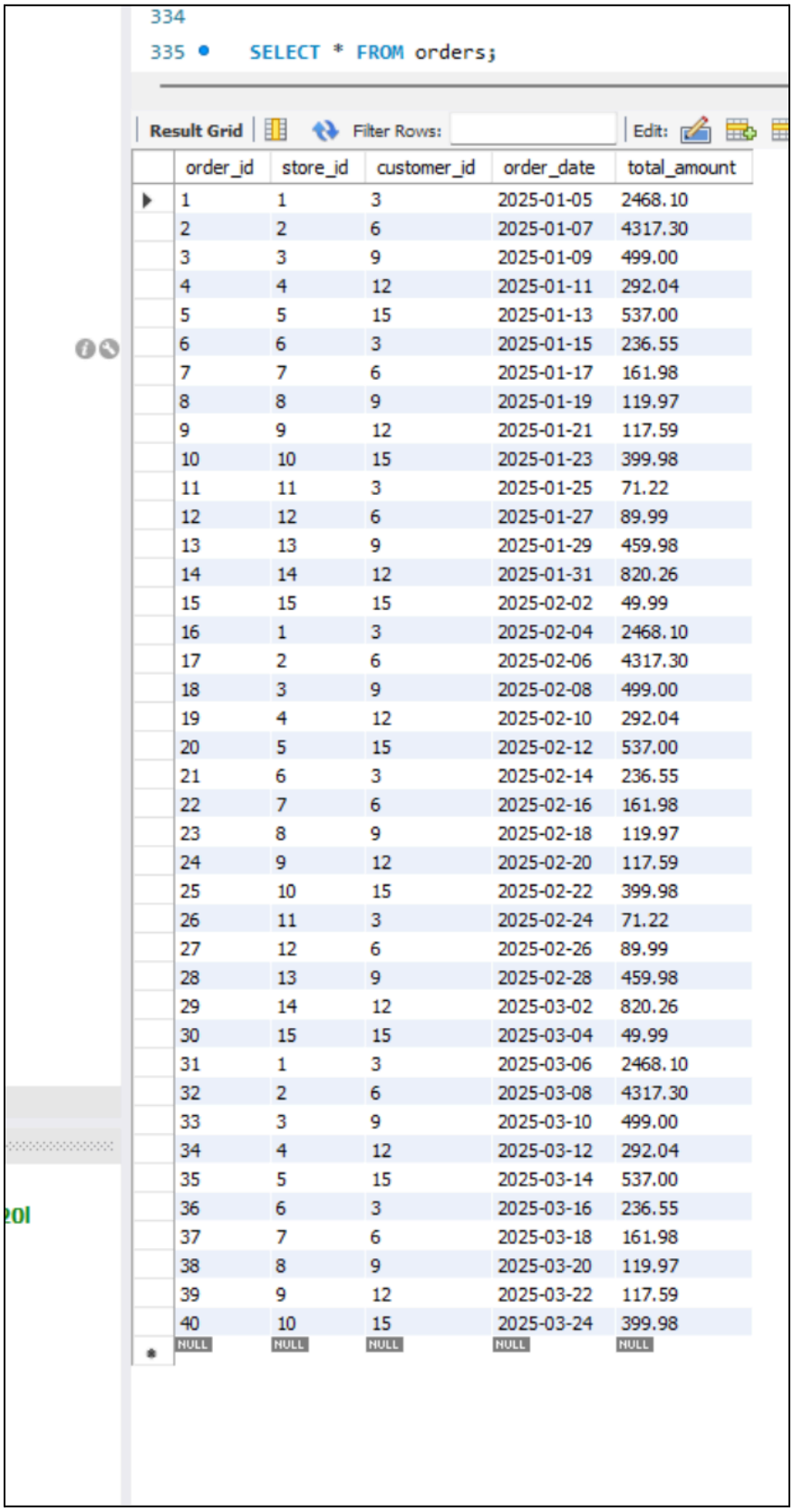
Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	supplier_id	category_id	warranty_id	product_name	product_price	product_description
1	3	1	2	AuroraBook 14 Laptop	1299.00	14-inch lightweight laptop with fast SSD storage…
2	3	1	3	AuroraBook 16 Laptop	1599.00	16-inch performance laptop for business users …
3	5	2	1	NovaTab 11 Tablet	499.00	Compact tablet for streaming, browsing, and no…
4	15	3	9	PulseBeat Wireless Headphones	149.00	Bluetooth over-ear headphones with active nois…
5	4	4	10	HaloLink Mesh Router	179.00	Dual-band mesh router kit for strong whole-hom…
6	14	5	12	PrismView 27 Monitor	249.00	27-inch IPS monitor with crisp color and adjusta…
7	6	6	14	ForgeKey Mechanical Keyboard	89.99	Mechanical keyboard with tactile switches and p…
8	6	6	14	GlidePro Wireless Mouse	39.99	Ergonomic wireless mouse with silent clicks and l…
9	2	7	11	TitanVault 1TB SSD	119.99	High-speed solid-state drive for laptops, deskto…
10	7	8	6	OrbitFit Smartwatch	199.99	Fitness smartwatch with heart-rate tracking an…
11	7	10	6	LumaAir Smart Plug	24.99	Connected plug for scheduling lamps, fans, and…
12	8	9	8	BreezeLine Coffee Maker	99.99	Programmable coffee maker with auto shutoff a…
13	10	9	8	FreshWave Air Purifier	229.99	Quiet air purifier designed for bedrooms and me…
14	1	11	5	ClearPrint Laser Printer	279.00	Compact monochrome printer for offices and ho…
15	11	14	15	PowerCell 20K Power Bank	49.99	Portable 20000mAh power bank with USB-C fas…
NULL	NULL	NULL	NULL	NULL	NULL	NULL

orders:

```
INSERT INTO `orders` (`order_id`, `store_id`, `customer_id`, `order_date`, `total_amount`) VALUES
(1, 1, 3, '2025-01-05', 2468.10),
(2, 2, 6, '2025-01-07', 4317.30),
(3, 3, 9, '2025-01-09', 499.00),
(4, 4, 12, '2025-01-11', 292.04),
(5, 5, 15, '2025-01-13', 537.00),
(6, 6, 3, '2025-01-15', 236.55),
(7, 7, 6, '2025-01-17', 161.98),
(8, 8, 9, '2025-01-19', 119.97),
(9, 9, 12, '2025-01-21', 117.59),
(10, 10, 15, '2025-01-23', 399.98),
(11, 11, 3, '2025-01-25', 71.22),
(12, 12, 6, '2025-01-27', 89.99),
(13, 13, 9, '2025-01-29', 459.98),
(14, 14, 12, '2025-01-31', 820.26),
(15, 15, 15, '2025-02-02', 49.99),
(16, 1, 3, '2025-02-04', 2468.10),
(17, 2, 6, '2025-02-06', 4317.30),
(18, 3, 9, '2025-02-08', 499.00),
(19, 4, 12, '2025-02-10', 292.04),
(20, 5, 15, '2025-02-12', 537.00),
(21, 6, 3, '2025-02-14', 236.55),
(22, 7, 6, '2025-02-16', 161.98),
(23, 8, 9, '2025-02-18', 119.97),
(24, 9, 12, '2025-02-20', 117.59),
(25, 10, 15, '2025-02-22', 399.98),
(26, 11, 3, '2025-02-24', 71.22),
(27, 12, 6, '2025-02-26', 89.99),
(28, 13, 9, '2025-02-28', 459.98),
(29, 14, 12, '2025-03-02', 820.26),
(30, 15, 15, '2025-03-04', 49.99),
(31, 1, 3, '2025-03-06', 2468.10),
(32, 2, 6, '2025-03-08', 4317.30),
(33, 3, 9, '2025-03-10', 499.00),
(34, 4, 12, '2025-03-12', 292.04),
(35, 5, 15, '2025-03-14', 537.00),
(36, 6, 3, '2025-03-16', 236.55),
(37, 7, 6, '2025-03-18', 161.98),
(38, 8, 9, '2025-03-20', 119.97),
(39, 9, 12, '2025-03-22', 117.59),
(40, 10, 15, '2025-03-24', 399.98);
```

orders (continued):



```
334
335 •   SELECT * FROM orders;
```

Result Grid | Filter Rows: | Edit:

order_id	store_id	customer_id	order_date	total_amount
1	1	3	2025-01-05	2468.10
2	2	6	2025-01-07	4317.30
3	3	9	2025-01-09	499.00
4	4	12	2025-01-11	292.04
5	5	15	2025-01-13	537.00
6	6	3	2025-01-15	236.55
7	7	6	2025-01-17	161.98
8	8	9	2025-01-19	119.97
9	9	12	2025-01-21	117.59
10	10	15	2025-01-23	399.98
11	11	3	2025-01-25	71.22
12	12	6	2025-01-27	89.99
13	13	9	2025-01-29	459.98
14	14	12	2025-01-31	820.26
15	15	15	2025-02-02	49.99
16	1	3	2025-02-04	2468.10
17	2	6	2025-02-06	4317.30
18	3	9	2025-02-08	499.00
19	4	12	2025-02-10	292.04
20	5	15	2025-02-12	537.00
21	6	3	2025-02-14	236.55
22	7	6	2025-02-16	161.98
23	8	9	2025-02-18	119.97
24	9	12	2025-02-20	117.59
25	10	15	2025-02-22	399.98
26	11	3	2025-02-24	71.22
27	12	6	2025-02-26	89.99
28	13	9	2025-02-28	459.98
29	14	12	2025-03-02	820.26
30	15	15	2025-03-04	49.99
31	1	3	2025-03-06	2468.10
32	2	6	2025-03-08	4317.30
33	3	9	2025-03-10	499.00
34	4	12	2025-03-12	292.04
35	5	15	2025-03-14	537.00
36	6	3	2025-03-16	236.55
37	7	6	2025-03-18	161.98
38	8	9	2025-03-20	119.97
39	9	12	2025-03-22	117.59
40	10	15	2025-03-24	399.98
NULL	NULL	NULL	NULL	NULL

order_data:

```sql
INSERT INTO `order_data` (`order_id`, `product_id`, `quantity_ordered`, `unit_price_at_sale`) VALUES
(1, 1, 2, 1234.05),
(2, 2, 3, 1439.10),
(3, 3, 1, 499.00),
(4, 4, 2, 146.02),
(5, 5, 3, 179.00),
(6, 6, 1, 236.55),
(7, 7, 2, 80.99),
(8, 8, 3, 39.99),
(9, 9, 1, 117.59),
(10, 10, 2, 199.99),
(11, 11, 3, 23.74),
(12, 12, 1, 89.99),
(13, 13, 2, 229.99),
(14, 14, 3, 273.42),
(15, 15, 1, 49.99),
(16, 1, 2, 1234.05),
(17, 2, 3, 1439.10),
(18, 3, 1, 499.00),
(19, 4, 2, 146.02),
(20, 5, 3, 179.00),
(21, 6, 1, 236.55),
(22, 7, 2, 80.99),
(23, 8, 3, 39.99),
(24, 9, 1, 117.59),
(25, 10, 2, 199.99),
(26, 11, 3, 23.74),
(27, 12, 1, 89.99),
(28, 13, 2, 229.99),
(29, 14, 3, 273.42),
(30, 15, 1, 49.99),
(31, 1, 2, 1234.05),
(32, 2, 3, 1439.10),
(33, 3, 1, 499.00),
(34, 4, 2, 146.02),
(35, 5, 3, 179.00),
(36, 6, 1, 236.55),
(37, 7, 2, 80.99),
(38, 8, 3, 39.99),
(39, 9, 1, 117.59),
(40, 10, 2, 199.99);
```

order_data (continued):

```
379 •   SELECT * FROM order_data;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

order_id	product_id	quantity_ordered	unit_price_at_sale
1	1	2	1234.05
2	2	3	1439.10
3	3	1	499.00
4	4	2	146.02
5	5	3	179.00
6	6	1	236.55
7	7	2	80.99
8	8	3	39.99
9	9	1	117.59
10	10	2	199.99
11	11	3	23.74
12	12	1	89.99
13	13	2	229.99
14	14	3	273.42
15	15	1	49.99
16	1	2	1234.05
17	2	3	1439.10
18	3	1	499.00
19	4	2	146.02
20	5	3	179.00
21	6	1	236.55
22	7	2	80.99
23	8	3	39.99
24	9	1	117.59
25	10	2	199.99
26	11	3	23.74
27	12	1	89.99
28	13	2	229.99
29	14	3	273.42
30	15	1	49.99
31	1	2	1234.05
32	2	3	1439.10
33	3	1	499.00
34	4	2	146.02
35	5	3	179.00
36	6	1	236.55
37	7	2	80.99
38	8	3	39.99
39	9	1	117.59
40	10	2	199.99
NULL	NULL	NULL	NULL

product_return:

```
INSERT INTO `product_return` (`return_id`, `order_id`, `product_id`, `quantity_returned`, `return_date`, `refund_amount`) VALUES
(1, 1, 1, 1, '2025-01-09', 1234.05),
(2, 2, 2, 1, '2025-01-12', 1439.10),
(3, 3, 3, 1, '2025-01-15', 499.00),
(4, 4, 4, 2, '2025-01-18', 292.04),
(5, 5, 5, 1, '2025-01-21', 179.00),
(6, 6, 6, 1, '2025-01-24', 236.55),
(7, 7, 7, 1, '2025-01-27', 80.99),
(8, 8, 8, 3, '2025-01-30', 119.97),
(9, 9, 9, 1, '2025-02-02', 117.59),
(10, 10, 10, 1, '2025-02-05', 199.99),
(11, 11, 11, 1, '2025-02-08', 23.74),
(12, 12, 12, 1, '2025-01-30', 89.99),
(13, 13, 13, 1, '2025-02-02', 229.99),
(14, 14, 14, 1, '2025-02-05', 273.42),
(15, 15, 15, 1, '2025-02-08', 49.99),
(16, 16, 1, 2, '2025-02-11', 2468.10),
(17, 17, 2, 1, '2025-02-14', 1439.10),
(18, 18, 3, 1, '2025-02-17', 499.00),
(19, 19, 4, 1, '2025-02-20', 146.02),
(20, 20, 5, 3, '2025-02-23', 537.00),
(21, 21, 6, 1, '2025-02-26', 236.55),
(22, 22, 7, 1, '2025-03-01', 80.99),
(23, 23, 8, 1, '2025-03-04', 39.99),
(24, 24, 9, 1, '2025-02-23', 117.59),
(25, 25, 10, 1, '2025-02-26', 199.99),
(26, 26, 11, 1, '2025-03-01', 23.74),
(27, 27, 12, 1, '2025-03-04', 89.99),
(28, 28, 13, 2, '2025-03-07', 459.98),
(29, 29, 14, 1, '2025-03-10', 273.42),
(30, 30, 15, 1, '2025-03-13', 49.99),
(31, 31, 1, 1, '2025-03-16', 1234.05),
(32, 32, 2, 3, '2025-03-19', 4317.30),
(33, 33, 3, 1, '2025-03-22', 499.00),
(34, 34, 4, 1, '2025-03-25', 146.02),
(35, 35, 5, 1, '2025-03-28', 179.00),
(36, 36, 6, 1, '2025-03-19', 236.55),
(37, 37, 7, 1, '2025-03-22', 80.99),
(38, 38, 8, 1, '2025-03-25', 39.99),
(39, 39, 9, 1, '2025-03-28', 117.59),
(40, 40, 10, 2, '2025-03-31', 399.98);
```

product_return (continued):

```
423 •   SELECT * FROM product_return;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

return_id	order_id	product_id	quantity_returned	return_date	refund_amount
1	1	1	1	2025-01-09	1234.05
2	2	2	1	2025-01-12	1439.10
3	3	3	1	2025-01-15	499.00
4	4	4	2	2025-01-18	292.04
5	5	5	1	2025-01-21	179.00
6	6	6	1	2025-01-24	236.55
7	7	7	1	2025-01-27	80.99
8	8	8	3	2025-01-30	119.97
9	9	9	1	2025-02-02	117.59
10	10	10	1	2025-02-05	199.99
11	11	11	1	2025-02-08	23.74
12	12	12	1	2025-01-30	89.99
13	13	13	1	2025-02-02	229.99
14	14	14	1	2025-02-05	273.42
15	15	15	1	2025-02-08	49.99
16	16	1	2	2025-02-11	2468.10
17	17	2	1	2025-02-14	1439.10
18	18	3	1	2025-02-17	499.00
19	19	4	1	2025-02-20	146.02
20	20	5	3	2025-02-23	537.00
21	21	6	1	2025-02-26	236.55
22	22	7	1	2025-03-01	80.99
23	23	8	1	2025-03-04	39.99
24	24	9	1	2025-02-23	117.59
25	25	10	1	2025-02-26	199.99
26	26	11	1	2025-03-01	23.74
27	27	12	1	2025-03-04	89.99
28	28	13	2	2025-03-07	459.98
29	29	14	1	2025-03-10	273.42
30	30	15	1	2025-03-13	49.99
31	31	1	1	2025-03-16	1234.05
32	32	2	3	2025-03-19	4317.30
33	33	3	1	2025-03-22	499.00
34	34	4	1	2025-03-25	146.02
35	35	5	1	2025-03-28	179.00
36	36	6	1	2025-03-19	236.55
37	37	7	1	2025-03-22	80.99
38	38	8	1	2025-03-25	39.99
39	39	9	1	2025-03-28	117.59
40	40	10	2	2025-03-31	399.98
NULL	NULL	NULL	NULL	NULL	NULL

service:

```sql
INSERT INTO `service` (`service_id`, `service_category`, `service_date`, `customer_id`, `store_id`, `employee_id`) VALUES
(1, 'Setup', '2025-01-08', 3, 1, 1),
(2, 'Diagnostics', '2025-01-11', 6, 2, 2),
(3, 'Repair', '2025-01-14', 9, 3, 3),
(4, 'Installation', '2025-01-17', 12, 4, 4),
(5, 'Warranty Claim', '2025-01-15', 15, 5, 5),
(6, 'Data Transfer', '2025-01-18', 3, 6, 6),
(7, 'Software Support', '2025-01-21', 6, 7, 7),
(8, 'Cleaning', '2025-01-24', 9, 8, 8),
(9, 'Calibration', '2025-01-27', 12, 9, 9),
(10, 'Upgrade Consultation', '2025-01-25', 15, 10, 10),
(11, 'Setup', '2025-01-28', 3, 11, 11),
(12, 'Diagnostics', '2025-01-31', 6, 12, 12),
(13, 'Repair', '2025-02-03', 9, 13, 13),
(14, 'Installation', '2025-02-06', 12, 14, 14),
(15, 'Warranty Claim', '2025-02-04', 15, 15, 15),
(16, 'Data Transfer', '2025-02-07', 3, 1, 1),
(17, 'Software Support', '2025-02-10', 6, 2, 2),
(18, 'Cleaning', '2025-02-13', 9, 3, 3),
(19, 'Calibration', '2025-02-16', 12, 4, 4),
(20, 'Upgrade Consultation', '2025-02-14', 15, 5, 5),
(21, 'Setup', '2025-02-17', 3, 6, 6),
(22, 'Diagnostics', '2025-02-20', 6, 7, 7),
(23, 'Repair', '2025-02-23', 9, 8, 8),
(24, 'Installation', '2025-02-26', 12, 9, 9),
(25, 'Warranty Claim', '2025-02-24', 15, 10, 10),
(26, 'Data Transfer', '2025-02-27', 3, 11, 11),
(27, 'Software Support', '2025-03-02', 6, 12, 12),
(28, 'Cleaning', '2025-03-05', 9, 13, 13),
(29, 'Calibration', '2025-03-08', 12, 14, 14),
(30, 'Upgrade Consultation', '2025-03-06', 15, 15, 15),
(31, 'Setup', '2025-03-09', 3, 1, 1),
(32, 'Diagnostics', '2025-03-12', 6, 2, 2),
(33, 'Repair', '2025-03-15', 9, 3, 3),
(34, 'Installation', '2025-03-18', 12, 4, 4),
(35, 'Warranty Claim', '2025-03-16', 15, 5, 5),
(36, 'Data Transfer', '2025-03-19', 3, 6, 6),
(37, 'Software Support', '2025-03-22', 6, 7, 7),
(38, 'Cleaning', '2025-03-25', 9, 8, 8),
(39, 'Calibration', '2025-03-28', 12, 9, 9),
(40, 'Upgrade Consultation', '2025-03-26', 15, 10, 10);
```

service (continued):

```
466
467 •   SELECT * FROM service;
468
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Co

service_id	service_category	service_date	customer_id	store_id	employee_id
1	Setup	2025-01-08	3	1	1
2	Diagnostics	2025-01-11	6	2	2
3	Repair	2025-01-14	9	3	3
4	Installation	2025-01-17	12	4	4
5	Warranty Claim	2025-01-15	15	5	5
6	Data Transfer	2025-01-18	3	6	6
7	Software Support	2025-01-21	6	7	7
8	Cleaning	2025-01-24	9	8	8
9	Calibration	2025-01-27	12	9	9
10	Upgrade Consultation	2025-01-25	15	10	10
11	Setup	2025-01-28	3	11	11
12	Diagnostics	2025-01-31	6	12	12
13	Repair	2025-02-03	9	13	13
14	Installation	2025-02-06	12	14	14
15	Warranty Claim	2025-02-04	15	15	15
16	Data Transfer	2025-02-07	3	1	1
17	Software Support	2025-02-10	6	2	2
18	Cleaning	2025-02-13	9	3	3
19	Calibration	2025-02-16	12	4	4
20	Upgrade Consultation	2025-02-14	15	5	5
21	Setup	2025-02-17	3	6	6
22	Diagnostics	2025-02-20	6	7	7
23	Repair	2025-02-23	9	8	8
24	Installation	2025-02-26	12	9	9
25	Warranty Claim	2025-02-24	15	10	10
26	Data Transfer	2025-02-27	3	11	11
27	Software Support	2025-03-02	6	12	12
28	Cleaning	2025-03-05	9	13	13
29	Calibration	2025-03-08	12	14	14
30	Upgrade Consultation	2025-03-06	15	15	15
31	Setup	2025-03-09	3	1	1
32	Diagnostics	2025-03-12	6	2	2
33	Repair	2025-03-15	9	3	3
34	Installation	2025-03-18	12	4	4
35	Warranty Claim	2025-03-16	15	5	5
36	Data Transfer	2025-03-19	3	6	6
37	Software Support	2025-03-22	6	7	7
38	Cleaning	2025-03-25	9	8	8
39	Calibration	2025-03-28	12	9	9
40	Upgrade Consultation	2025-03-26	15	10	10
NULL	NULL	NULL	NULL	NULL	NULL

service 1 ×

Queries:

Store sales report showing units sold, total orders, and gross sales per store

```
473     #Store sales report showing units sold, total orders, and gross sales per store
474 •   SELECT
475         s.store_id,
476         s.store_name,
477         COUNT(DISTINCT o.order_id) AS total_orders,
478         SUM(od.quantity_ordered) AS total_units_sold,
479         SUM(od.quantity_ordered * od.unit_price_at_sale) AS gross_sales
480     FROM store s
481     JOIN orders o ON o.store_id = s.store_id
482     JOIN order_data od ON od.order_id = o.order_id
483     GROUP BY s.store_id, s.store_name
484     ORDER BY gross_sales DESC;
```

100% 27:484

Result Grid Filter Rows: Search Export:

store_id	store_name	total_orders	total_units_s...	gross_sales
2	Lakeside Retail Hub	3	9	12951.90
1	Northgate Tech Center	3	6	7404.30
14	Union Square Tech	2	6	1640.52
5	Midtown Market Tech	3	9	1611.00
3	Downtown Device Store	3	3	1497.00
10	Prairie View Tech	3	6	1199.94
13	Hillcrest Plaza	2	4	919.96
4	Riverfront Electronics	3	6	876.12
6	Westfield Commons	3	3	709.65
7	Harbor Point Retail	3	6	485.94
8	Oakwood Square	3	9	359.91
9	Summit Crossing	3	3	352.77
12	Pinecrest Outlet	2	2	179.98
11	Belmont Station	2	6	142.44
15	Cedar Park Electronics	2	2	99.98

Result 4

Customer order history report showing customer purchases over time

customer_id	first_name	last_name	order_id	order_date	total_amou...
3	Maya	Patel	1	2025-01-05	2468.10
3	Maya	Patel	6	2025-01-15	236.55
3	Maya	Patel	11	2025-01-25	71.22
3	Maya	Patel	16	2025-02-04	2468.10
3	Maya	Patel	21	2025-02-14	236.55
3	Maya	Patel	26	2025-02-24	71.22
3	Maya	Patel	31	2025-03-06	2468.10
3	Maya	Patel	36	2025-03-16	236.55
6	Liam	Mitchell	2	2025-01-07	4317.30
6	Liam	Mitchell	7	2025-01-17	161.98
6	Liam	Mitchell	12	2025-01-27	89.99
6	Liam	Mitchell	17	2025-02-06	4317.30
6	Liam	Mitchell	22	2025-02-16	161.98
6	Liam	Mitchell	27	2025-02-26	89.99
6	Liam	Mitchell	32	2025-03-08	4317.30
6	Liam	Mitchell	37	2025-03-18	161.98
9	Emma	Walker	3	2025-01-09	499.00
9	Emma	Walker	8	2025-01-19	119.97
9	Emma	Walker	13	2025-01-29	459.98
9	Emma	Walker	18	2025-02-08	499.00
9	Emma	Walker	23	2025-02-18	119.97
9	Emma	Walker	28	2025-02-28	459.98
9	Emma	Walker	33	2025-03-10	499.00
9	Emma	Walker	38	2025-03-20	119.97
12	Mason	Price	4	2025-01-11	292.04
12	Mason	Price	9	2025-01-21	117.59
12	Mason	Price	14	2025-01-31	820.26
12	Mason	Price	19	2025-02-10	292.04
12	Mason	Price	24	2025-02-20	117.59
12	Mason	Price	29	2025-03-02	820.26
12	Mason	Price	34	2025-03-12	292.04
12	Mason	Price	39	2025-03-22	117.59
15	Harper	Ward	5	2025-01-13	537.00
15	Harper	Ward	10	2025-01-23	399.98
15	Harper	Ward	15	2025-02-02	49.99
15	Harper	Ward	20	2025-02-12	537.00
15	Harper	Ward	25	2025-02-22	399.98
15	Harper	Ward	30	2025-03-04	49.99
15	Harper	Ward	35	2025-03-14	537.00
15	Harper	Ward	40	2025-03-24	399.98

```sql
#Customer order history report showing customer purchases over time
SELECT
    c.customer_id,
    c.first_name,
    c.last_name,
    o.order_id,
    o.order_date,
    o.total_amount
FROM customer c
JOIN orders o ON o.customer_id = c.customer_id
ORDER BY c.customer_id, o.order_date;
```

Customer return report showing returned products, return dates, and refund amounts

customer_id	first_name	last_name	return_id	return_date	product_name	quantity_return...	refund_amou...
3	Maya	Patel	1	2025-01-09	AuroraBook 14 Laptop	1	1234.05
6	Liam	Mitchell	2	2025-01-12	AuroraBook 16 Laptop	1	1439.10
9	Emma	Walker	3	2025-01-15	NovaTab 11 Tablet	1	499.00
12	Mason	Price	4	2025-01-18	PulseBeat Wireless Headphones	2	292.04
15	Harper	Ward	5	2025-01-21	HaloLink Mesh Router	1	179.00
3	Maya	Patel	6	2025-01-24	PrismView 27 Monitor	1	236.55
6	Liam	Mitchell	7	2025-01-27	ForgeKey Mechanical Keyboard	1	80.99
9	Emma	Walker	8	2025-01-30	GlidePro Wireless Mouse	3	119.97
6	Liam	Mitchell	12	2025-01-30	BreezeLine Coffee Maker	1	89.99
12	Mason	Price	9	2025-02-02	TitanVault 1TB SSD	1	117.59
9	Emma	Walker	13	2025-02-02	FreshWave Air Purifier	1	229.99
15	Harper	Ward	10	2025-02-05	OrbitFit Smartwatch	1	199.99
12	Mason	Price	14	2025-02-05	ClearPrint Laser Printer	1	273.42
15	Harper	Ward	15	2025-02-08	PowerCell 20K Power Bank	1	49.99
3	Maya	Patel	11	2025-02-08	LumaAir Smart Plug	1	23.74
3	Maya	Patel	16	2025-02-11	AuroraBook 14 Laptop	2	2468.10
6	Liam	Mitchell	17	2025-02-14	AuroraBook 16 Laptop	1	1439.10
9	Emma	Walker	18	2025-02-17	NovaTab 11 Tablet	1	499.00
12	Mason	Price	19	2025-02-20	PulseBeat Wireless Headphones	1	146.02
15	Harper	Ward	20	2025-02-23	HaloLink Mesh Router	3	537.00
12	Mason	Price	24	2025-02-23	TitanVault 1TB SSD	1	117.59
15	Harper	Ward	25	2025-02-26	OrbitFit Smartwatch	1	199.99
3	Maya	Patel	21	2025-02-26	PrismView 27 Monitor	1	236.55
3	Maya	Patel	26	2025-03-01	LumaAir Smart Plug	1	23.74
6	Liam	Mitchell	22	2025-03-01	ForgeKey Mechanical Keyboard	1	80.99
9	Emma	Walker	23	2025-03-04	GlidePro Wireless Mouse	1	39.99
6	Liam	Mitchell	27	2025-03-04	BreezeLine Coffee Maker	1	89.99
9	Emma	Walker	28	2025-03-07	FreshWave Air Purifier	2	459.98
12	Mason	Price	29	2025-03-10	ClearPrint Laser Printer	1	273.42
15	Harper	Ward	30	2025-03-13	PowerCell 20K Power Bank	1	49.99
3	Maya	Patel	31	2025-03-16	AuroraBook 14 Laptop	1	1234.05
6	Liam	Mitchell	32	2025-03-19	AuroraBook 16 Laptop	3	4317.30
3	Maya	Patel	36	2025-03-19	PrismView 27 Monitor	1	236.55
9	Emma	Walker	33	2025-03-22	NovaTab 11 Tablet	1	499.00
6	Liam	Mitchell	37	2025-03-22	ForgeKey Mechanical Keyboard	1	80.99
9	Emma	Walker	38	2025-03-25	GlidePro Wireless Mouse	1	39.99
12	Mason	Price	34	2025-03-25	PulseBeat Wireless Headphones	1	146.02
12	Mason	Price	39	2025-03-28	TitanVault 1TB SSD	1	117.59
15	Harper	Ward	35	2025-03-28	HaloLink Mesh Router	1	179.00
15	Harper	Ward	40	2025-03-31	OrbitFit Smartwatch	2	399.98

Monthly sales trend report showing total orders and total sales by month

```
513
514     #Monthly sales trend report showing total orders and total sales by month
515     SELECT
516         YEAR(o.order_date) AS sales_year,
517         MONTH(o.order_date) AS sales_month,
518         COUNT(o.order_id) AS total_orders,
519         SUM(o.total_amount) AS monthly_sales
520     FROM orders o
521     GROUP BY YEAR(o.order_date), MONTH(o.order_date)
522     ORDER BY sales_year, sales_month;
523
```

100% 34:522

Result Grid Filter Rows: Search Export:

sales_year	sales_month	total_orders	monthly_sales
2025	1	14	10590.96
2025	2	14	9820.69
2025	3	12	10019.76

Product sales report showing units sold and total sales by product

```sql
#Product sales report showing units sold and total sales by product
SELECT
    p.product_id,
    p.product_name,
    SUM(od.quantity_ordered) AS total_units_sold,
    SUM(od.quantity_ordered * od.unit_price_at_sale) AS total_product_sales
FROM product p
JOIN order_data od ON od.product_id = p.product_id
GROUP BY p.product_id, p.product_name
ORDER BY total_product_sales DESC;
```

100% 35:533

Result Grid Filter Rows: Search Export:

product_...	product_name	total_units_s...	total_product_sa...
2	AuroraBook 16 Laptop	9	12951.90
1	AuroraBook 14 Laptop	6	7404.30
14	ClearPrint Laser Printer	6	1640.52
5	HaloLink Mesh Router	9	1611.00
3	NovaTab 11 Tablet	3	1497.00
10	OrbitFit Smartwatch	6	1199.94
13	FreshWave Air Purifier	4	919.96
4	PulseBeat Wireless Headphones	6	876.12
6	PrismView 27 Monitor	3	709.65
7	ForgeKey Mechanical Keyboard	6	485.94
8	GlidePro Wireless Mouse	9	359.91
9	TitanVault 1TB SSD	3	352.77
12	BreezeLine Coffee Maker	2	179.98
11	LumaAir Smart Plug	6	142.44
15	PowerCell 20K Power Bank	2	99.98

View 1:

```
535 •  CREATE VIEW customer_spending_summary AS
536    SELECT
537        c.customer_id,
538        c.first_name,
539        c.last_name,
540        COUNT(o.order_id) AS total_orders,
541        SUM(o.total_amount) AS total_spent,
542        AVG(o.total_amount) AS average_order_value
543    FROM customer c
544    JOIN orders o ON o.customer_id = c.customer_id
545    GROUP BY c.customer_id, c.first_name, c.last_name;
546 •  SELECT * FROM customer_spending_summary;
```

100% 41:546

Result Grid Filter Rows: Search Export:

customer_id	first_name	last_name	total_orders	total_spe...	average_order_val...
3	Maya	Patel	8	8256.39	1032.048750
6	Liam	Mitchell	8	13617.82	1702.227500
9	Emma	Walker	8	2776.87	347.108750
12	Mason	Price	8	2869.41	358.676250
15	Harper	Ward	8	2910.92	363.865000

View 2:

```
548 •  CREATE VIEW product_catalog_summary AS
549    SELECT
550        p.product_id,
551        p.product_name,
552        s.supplier_name,
553        c.category_name,
554        w.warranty_name,
555        p.product_price
556    FROM product p
557    JOIN supplier s ON p.supplier_id = s.supplier_id
558    JOIN category c ON p.category_id = c.category_id
559    JOIN warranty w ON p.warranty_id = w.warranty_id;
560 •  SELECT * FROM product_catalog_summary;
```

100% 39:560

Result Grid Filter Rows: Search Export:

product_...	product_name	supplier_name	category_name	warranty_name	product_pri...
1	AuroraBook 14 Laptop	BluePeak Electronics	Laptops	Extended Coverage	1299.00
2	AuroraBook 16 Laptop	BluePeak Electronics	Laptops	Premium Protection	1599.00
3	NovaTab 11 Tablet	Evergreen Consumer Supply	Tablets	Standard Coverage	499.00
4	PulseBeat Wireless Headphones	Urban Circuit Supply	Audio	Audio Care	149.00
5	HaloLink Mesh Router	Summit Tech Wholesale	Networking	Smart Home Care	179.00
6	PrismView 27 Monitor	Vertex Global Supply	Displays	Office Care	249.00
7	ForgeKey Mechanical Keyboard	MetroParts Trading	Peripherals	Mobile Care	89.99
8	GlidePro Wireless Mouse	MetroParts Trading	Peripherals	Mobile Care	39.99
9	TitanVault 1TB SSD	Northline Distributors	Storage	Storage Shield	119.99
10	OrbitFit Smartwatch	Horizon Appliance Supply	Wearables	Device Care Plus	199.99
11	LumaAir Smart Plug	Horizon Appliance Supply	Smart Home	Device Care Plus	24.99
12	BreezeLine Coffee Maker	Quantum Logistics Supply	Home Applian...	Appliance Care	99.99
13	FreshWave Air Purifier	Beacon Product Group	Home Applian...	Appliance Care	229.99
14	ClearPrint Laser Printer	Apex Components Ltd	Office Equipm...	Business Protection	279.00
15	PowerCell 20K Power Bank	Prime Source Solutions	Power Access...	Power Care	49.99

product_catalog_summary 12

Procedure 1:

```
562    DELIMITER //
563
564    CREATE PROCEDURE GetCustomerOrders(IN input_customer_id INT)
565    BEGIN
566        SELECT
567            c.customer_id,
568            c.first_name,
569            c.last_name,
570            o.order_id,
571            o.order_date,
572            o.total_amount
573        FROM customer c
574        JOIN orders o ON o.customer_id = c.customer_id
575        WHERE c.customer_id = input_customer_id
576        ORDER BY o.order_date;
577    END //
578
579    DELIMITER ;
580
```

100% 1:582

Result Grid Filter Rows: Search Export:

customer_id	first_name	last_name	order_id	order_date	total_amou...
3	Maya	Patel	1	2025-01-05	2468.10
3	Maya	Patel	6	2025-01-15	236.55
3	Maya	Patel	11	2025-01-25	71.22
3	Maya	Patel	16	2025-02-04	2468.10
3	Maya	Patel	21	2025-02-14	236.55
3	Maya	Patel	26	2025-02-24	71.22
3	Maya	Patel	31	2025-03-06	2468.10
3	Maya	Patel	36	2025-03-16	236.55

Procedure 2:

```
584    DELIMITER //
585
586    CREATE PROCEDURE GetProductReturns(IN input_product_id INT)
587    BEGIN
588        SELECT
589            pr.return_id,
590            pr.return_date,
591            p.product_name,
592            pr.quantity_returned,
593            pr.refund_amount
594        FROM product_return pr
595        JOIN product p ON p.product_id = pr.product_id
596        WHERE p.product_id = input_product_id
597        ORDER BY pr.return_date;
598    END //
599
600    DELIMITER ;
601
602    CALL GetProductReturns(1);
603
```

100% 27:602

Result Grid Filter Rows: Search Export:

return_id	return_date	product_name	quantity_return...	refund_amou...
1	2025-01-09	AuroraBook 14 Laptop	1	1234.05
16	2025-02-11	AuroraBook 14 Laptop	2	2468.10
31	2025-03-16	AuroraBook 14 Laptop	1	1234.05

Procedure 3:

```
606 •  CREATE PROCEDURE GetStoreServices(IN input_store_id INT)
607    BEGIN
608        SELECT
609            s.service_id,
610            s.service_category,
611            s.service_date,
612            s.customer_id,
613            s.employee_id,
614            st.store_name
615        FROM service s
616        JOIN store st ON st.store_id = s.store_id
617        WHERE s.store_id = input_store_id
618        ORDER BY s.service_date;
619    END //
620
621    DELIMITER ;
622
623 •  CALL GetStoreServices(1);
624
```

100% | 26:623

Result Grid Filter Rows: Search Export:

service_id	service_catego...	service_date	customer_id	employee_id	store_name
1	Setup	2025-01-08	3	1	Northgate Tech Center
16	Data Transfer	2025-02-07	3	1	Northgate Tech Center
31	Setup	2025-03-09	3	1	Northgate Tech Center